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palm
PalmSource
Conference 2000



Developing Hardware Add-ons on Palm OS® Licensee Devices

Steve McGuigan
Expansion Market Development Manager, Palm OS Platform

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Agenda

- Format
- Introduction of Presenters
  - Handspring
    - *Robert Gosselin: Manager, Developer Technical Support*
  - Palm, Inc.
    - *Frank Canova: Director, Palm Labs*
  - Sony
    - *Masaharu Yanaga: Senior General Manager*
  - TRG
    - *Mark Kubovich: VP of Operations*
- Q&A
- Proceed to Roundtable discussion 102

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Handspring

Robert Gosselin
Manager, Developer
Technical Support

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Springboard Expansion Slot

What Is It?

- Handspring's expansion technology
  - First true plug and play technology for hardware expansion on the handheld
- Bring hardware expansion to the handheld computer
- Agenda:
  - Technical details
  - Developer Programs
  - Marketing opportunities

Springboard Interface Details

Simple Parallel Bus

- Electrical interface:
  - 24 address / 16 Data lines,
  - Control lines: OE*, WE*, Int*, Reset*, 2 Chips Selects and 2 Card Detects
  - Flexible power management:
    - 100 mA @ 3.0–3.6V power is provided (VCC)
    - Optional charging voltage (VDOCK)
  - Microphone signals (MIC+/-)
- Memory mapped in CPU address range

Springboard Interface Details

Mechanically "Open"

- Slot is open on back and top to enable larger modules



Springboard Interface Details

First True Plug-and-Play Technology for Hardware Expansion on the Handheld

- As soon as it is inserted, module can be use:
  - Execution in place versus copying in internal memory
    - Setup application
    - Welcome application
- Standard Palm OS® file format:
  - No “special” file format

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Development Programs

No-Cost Program



- No-cost Program:
  - Access to all the documentation for development
  - Technical support: DevSupport@handspring.com
  - Marketing support: DevInfo@handspring.com
  - Program support: DevProgram@handspring.com
- Premier Program:
  - Everything the no-cost program gives +
  - “First in queue” for Technical Support
  - Up to 20 handhelds @ 20%
  - $6000/year

Support Programs



- Springboard Development Resources
  - Application-Only Springboard Modules
  - Springboard Connectors and Plastics
  - Ultrasonic Welding Services
  - Software and Hardware Technical Resources
  - Marketing and Distribution Partners
- Marcom and Print/Web Design Resources
- Packaging—Retail and E-Commerce Boxes
- International

www.handspring.com/developers/marketing_product.jhtml

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Marketing Opportunities



- Handspring Media Coverage and press release guideline
- Handspring Collateral
- eCommerce: Handspring Web store
- Channel Marketing
- National and international distribution
- Trade Show participation

Springboard Expansion Slot



- Design from the ground up for handheld expansion:
  - For the customer
  - For the developer
  - Basis for almost anything you want!
- Built with the Palm OS® philosophy in mind
- Free license for module design!
- First true plug-and-play technology for hardware expansion on the handheld



Palm, Inc.

Frank Canova
Director, Palm Labs

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Mission Statement

- Create a world class expansion architecture
- Grow the Palm Economy by providing expansion opportunities
- Enable developers by creating comprehensive support program for all phases
- Create marketing and distribution vehicles for your products

Why a Dual Expansion Architecture

- Allows for wearable devices with expansion
- Based on a robust industry standard
- Interoperable with other consumer devices
- I/O capable
- Fast transfer rate 10MB/sec
- USB capable









Dual Expansion Architecture

- Expansion card slot
  - Why we chose SD and MMC:
    - Size (Think: postage stamp)
    - I/O (for Bluetooth, Ethernet, modems, cameras, etc.)
    - Security (supports digital rights management)
    - Flexibility (used in cameras, video recorders, MP3's, etc.)
    - Open platform (no licensing fees = lots of third-party solutions)
    - Cost (less expensive media because of volume)



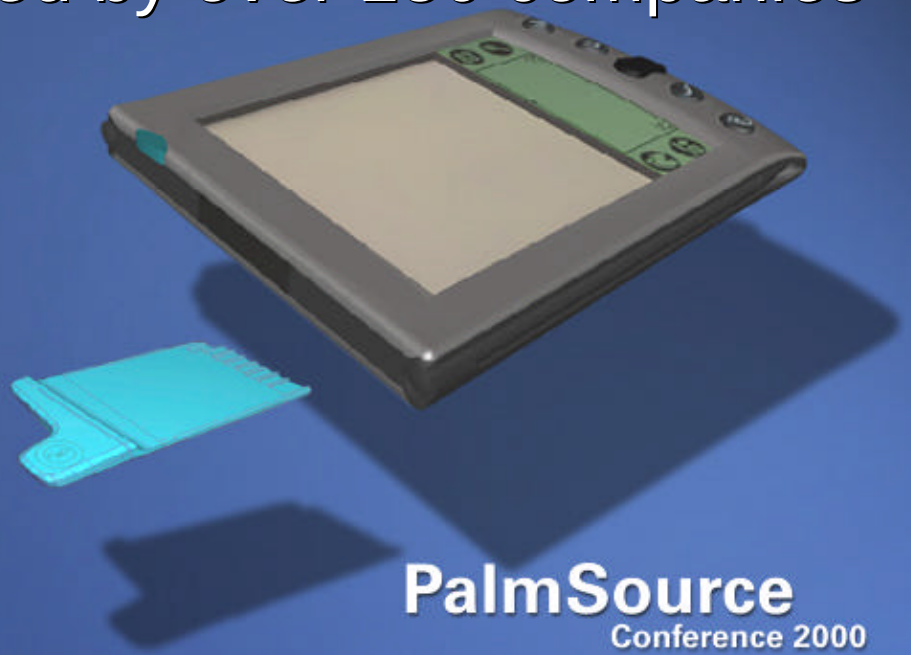
Dual Expansion Architecture

- Universal Connector
  - Create one product for Palm's future product family
  - Enables plug & play convenience
  - Can work in concert with SD or MMC cards
  - USB capable



Expansion Card Slot

- Supports SD, MMC and SDIO
- Interoperable between SD enabled products
- SDIO combination card: IO function + memory
- Industry standard backed by over 150 companies



SD Interoperability



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Removable Media Advantages

- SD Memory used for backup, distribution
- SDIO has many advantages
  - Open Standard
  - Fast—10MB/Sec (SCSI speeds)
  - Bluetooth and Modem standardized
  - Plug-and-Play standard
  - High register capacity (up to 128MB on card) allows for new class of content offerings on device
- Increase options for user
- Enabler of the Palm Economy

Comparison of Media Types

	SDCard	Memory Stick	CompactFlash	Springboard
Open Standard	YES (Supported by 153 Comapnies)	No	Yes	No
Data Transfer Speed (Max.)*	10MB/s	2.45 MB/s	2.2 MB/s	11 MB/s
SPI Support	Yes	No	No	No
SDMI Security	Version 2.0	Version 2.0 [1,2]	Planned	None
Volume/Size (cubic mm)	1,612mm	3,010mm	5,141mm	15,325mm
Write Protect	Physical	Physical	Electrical	None



Notes:
1. MagicGate on select higher capacities only
2. Not aligned with DVD Encryption Standard
*Also “Duo” version without I/O support.
*Theoretical maximum, rates may vary under certain conditions
SPI-Serial peripheral interface
Compact media illustrations not to scale or actual size

SD/MMC Co-Existence

Shared Marketplace



- MMC = Primarily “Non-secure” applications:
  - Digital Still and Video Cameras, HPCs
  - Many Cellular, GPS, E-book applications
  - SDMI 1.0 Compliant Audio Players
  - ROM versions for low-cost distribution

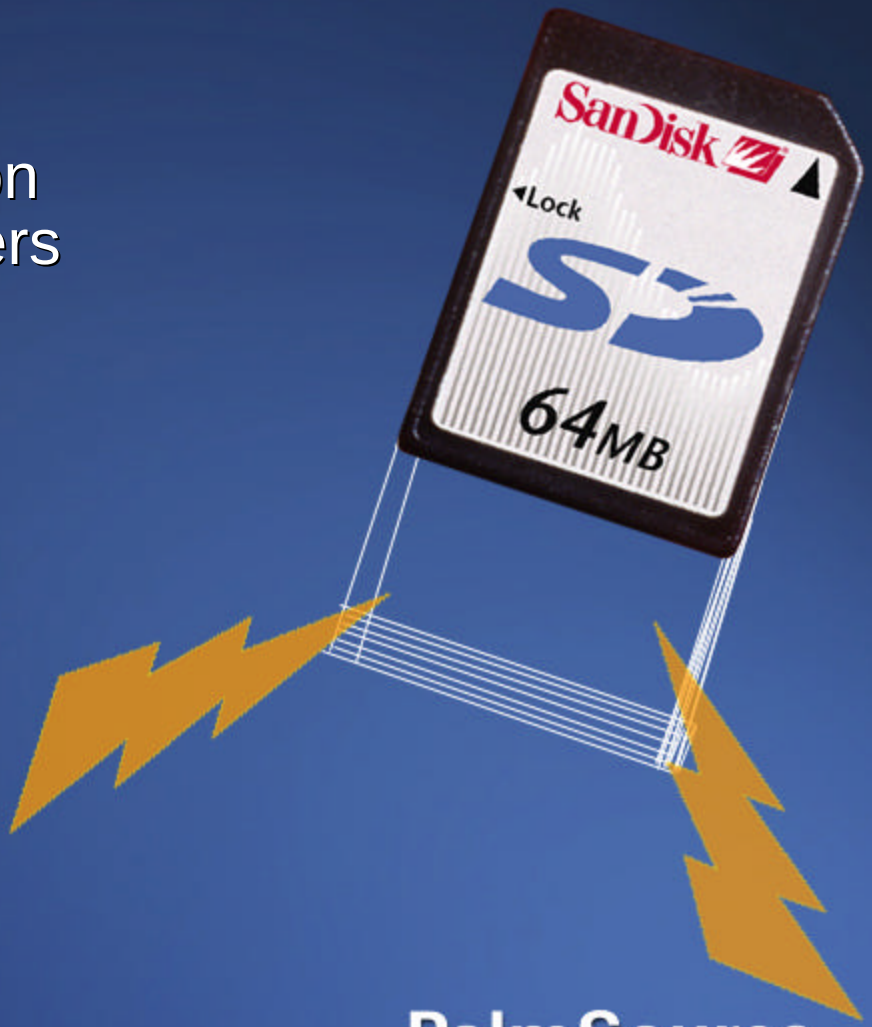
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- SD = Primarily “Secure” Audio/Video applications
  - SDMI 2.0 Compliant Audio Players
  - Additional Cellular, GPS and E-book apps
  - Future Personal Video Players—2002

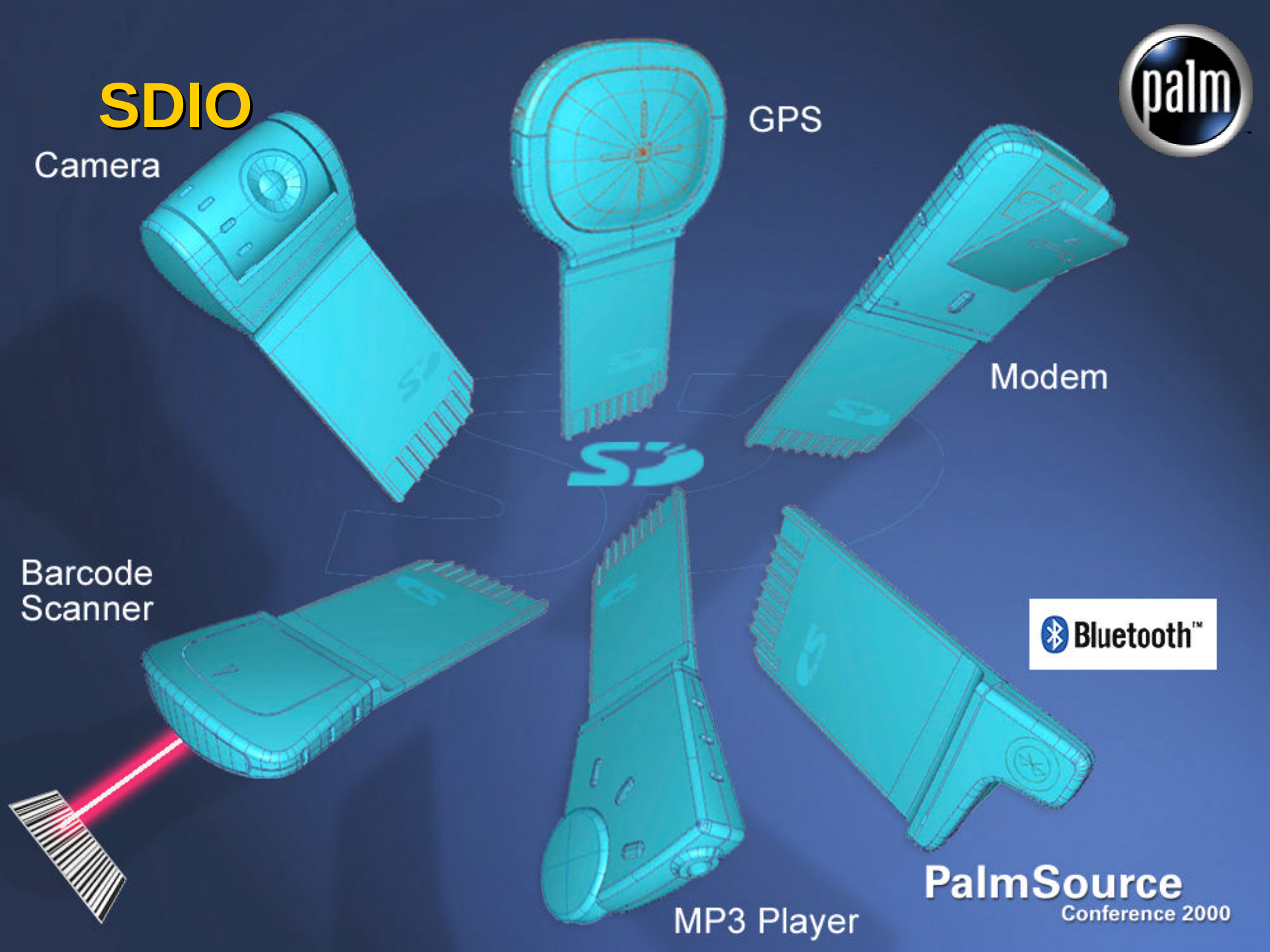


SDIO

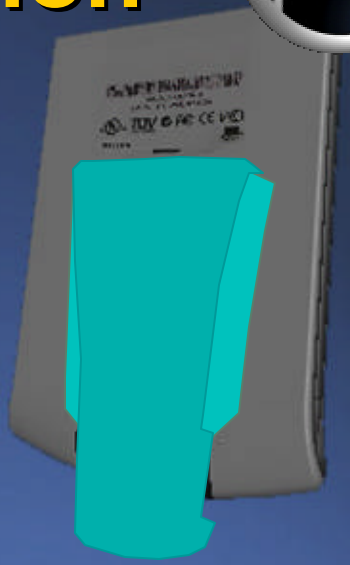
- Expandable SD Card
  - Alleviates slot contention for memory and I/O users
  - I/O only possible for cost-sensitive users
  - Full adherence to SDA specifications for I/O and combination card functionality
  - Maximum flexibility for users with multiple I/O needs



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SDIO
Camera
GPS
Modem
Barcode
Scanner
Bluetooth™
MP3 Player
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Universal Connector Expansion



- Common electrical and mechanical interface for peripherals
- Enables broad range of accessories over a large portfolio of Palm™ products
- Enables plug-and-play convenience
- USB connectivity for faster transfers
- RS232 connectivity to simplify interfacing
- Serial rates up to 230 Kbaud
- EDK ensures ease of development

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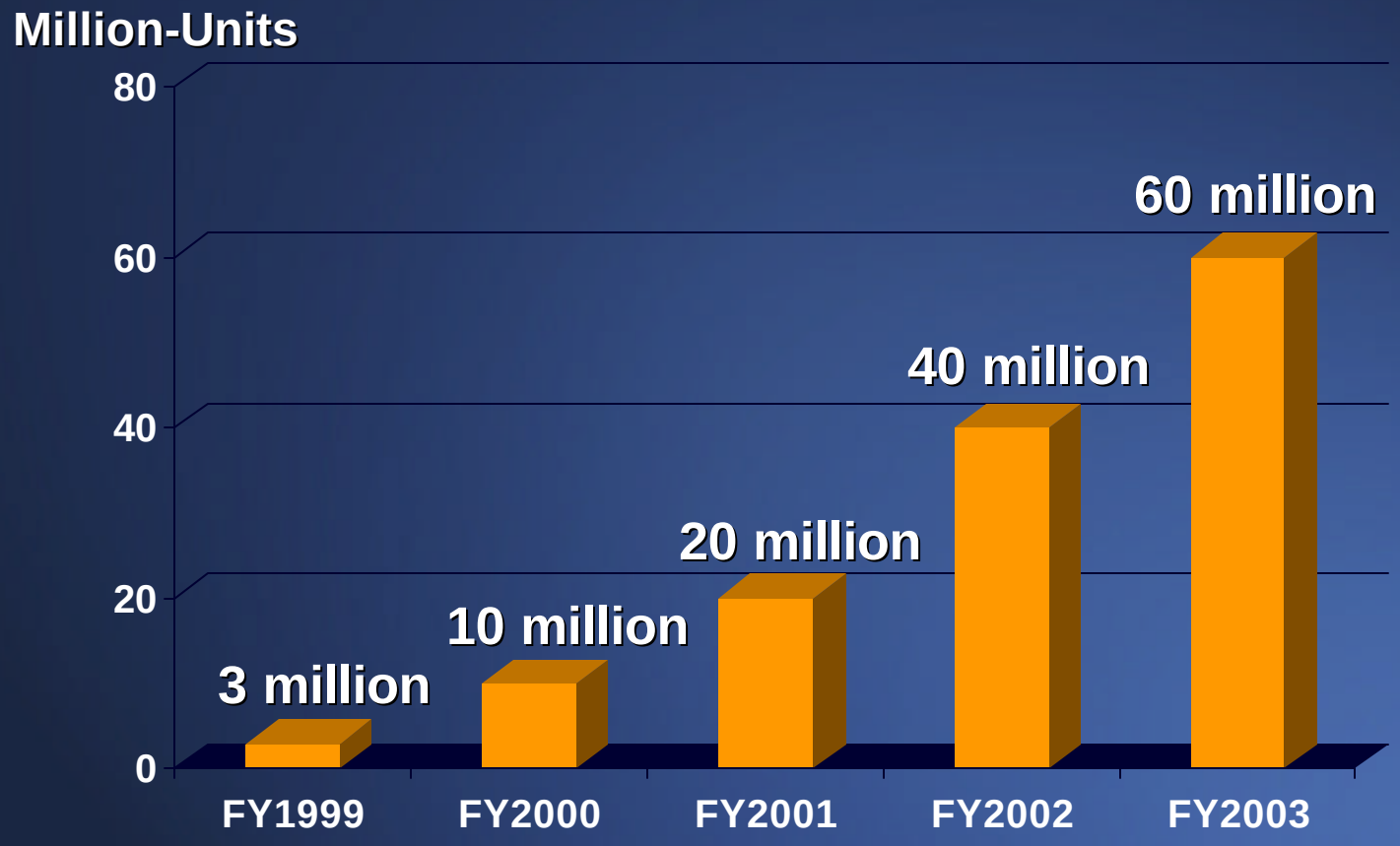
Sony

Masaharu Yanaga
Senior General Manager
Memory Stick Division

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Worldwide Shipment of Memory Stick® Products

SONY

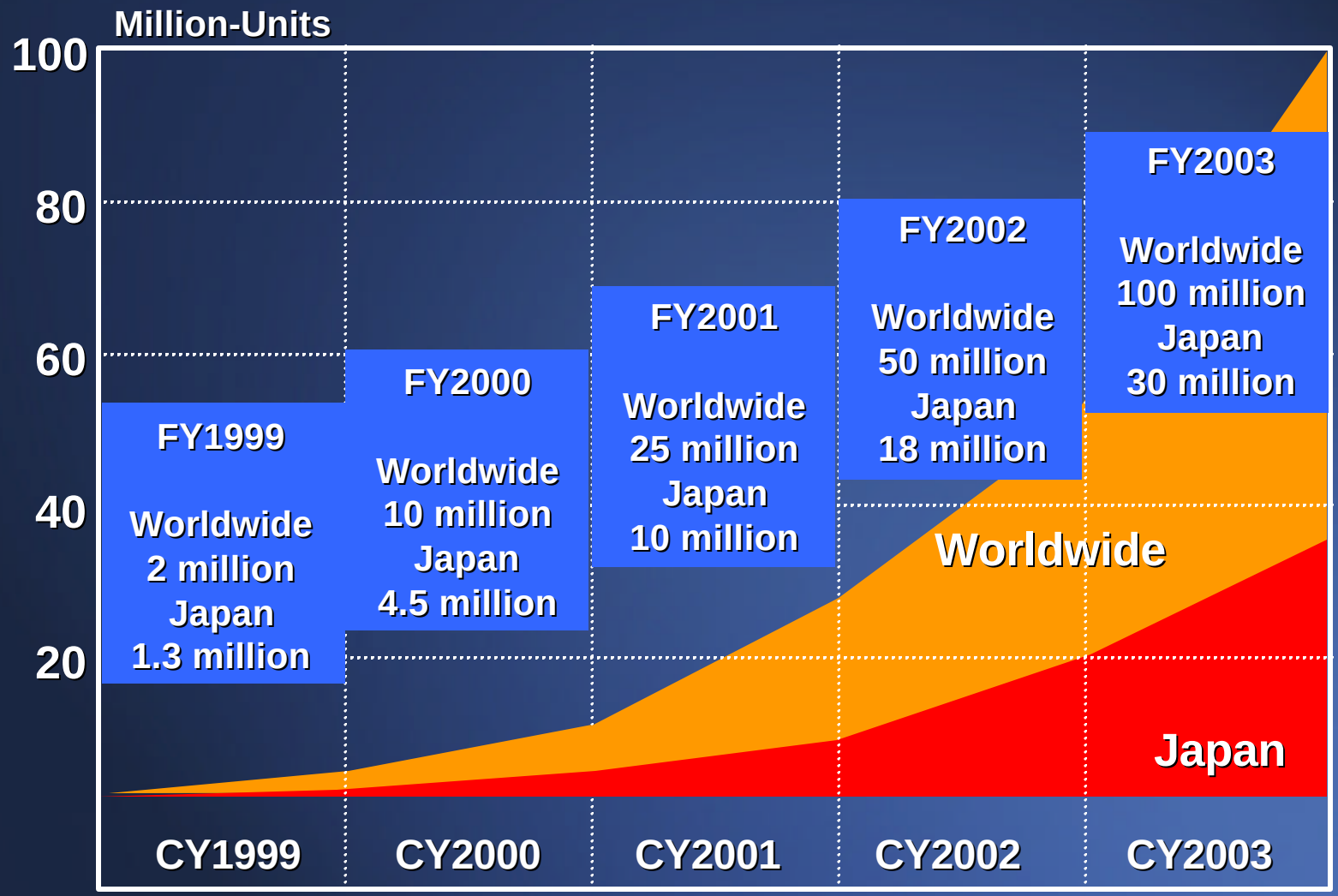


Sony Survey

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Cumulative Shipment of Memory Stick Compliant Products

SONY



Memory Stick Supporting Companies

SONY

Memory Stick-compatible Products Grow Day-by-day

- 20 Categories, 76 models available worldwide
- At Comdex…
  - 117 companies support Memory Stick products
  - 57 companies exhibited at “Memory Stick Pavilion”
    - Car Audio, Portable Audio, PDA, Printer, Scanner, DSC, Cellular, Media

MEMORY STICK

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Memory Stick Expansion Module

SONY



GPS Module





Bluetooth Module



Camera Module



Fingerprint Recognition Module

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Memory Stick Expansion Module

SONY

Conceptual “Tablet” with “GPS Stick” and Multiuse “Jog Stick”



Current Sony CLIÉ Handheld with “GPS Stick”



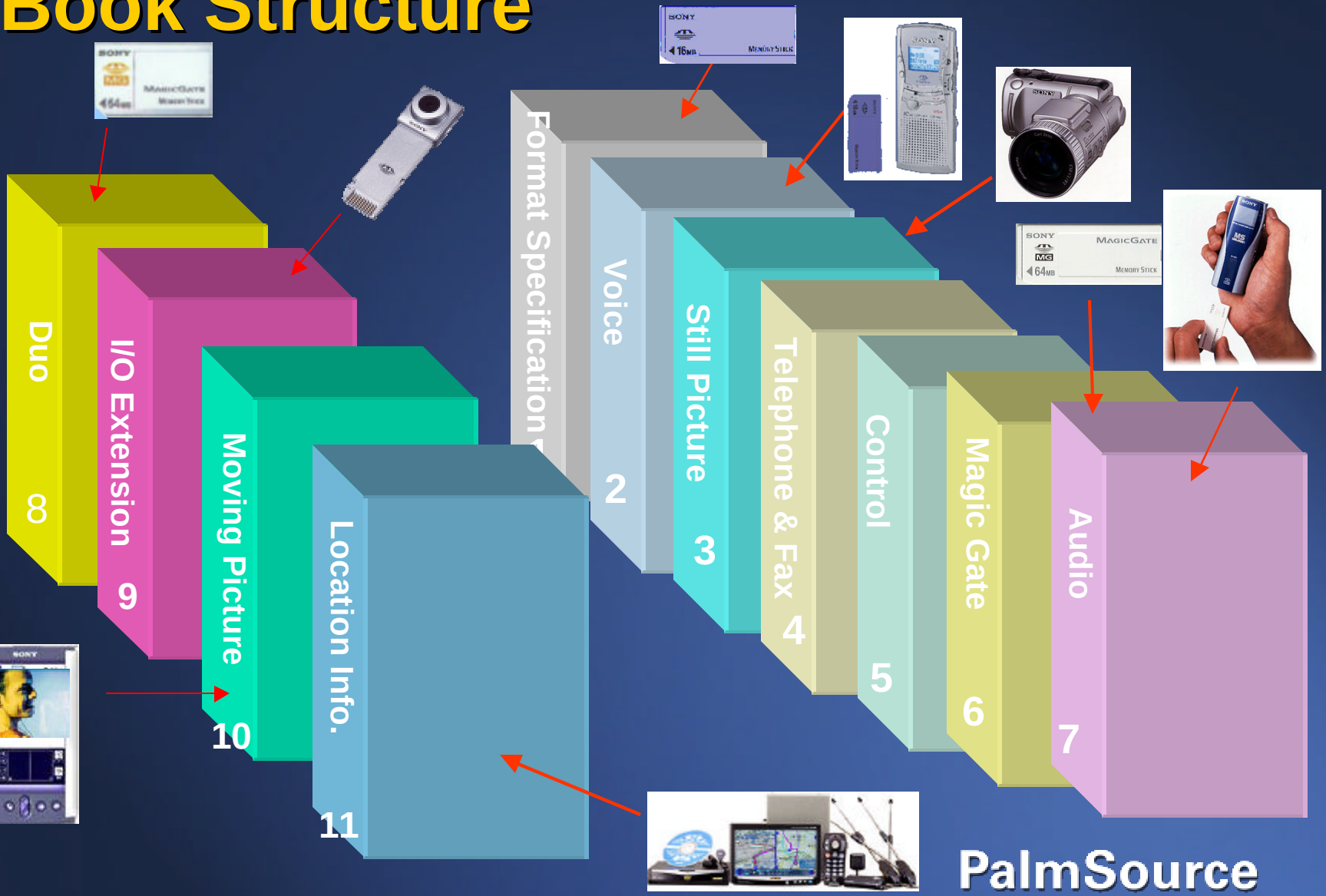
Digital Cellular with “Digital Still Camera Stick”



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Memory Stick Format Book Structure

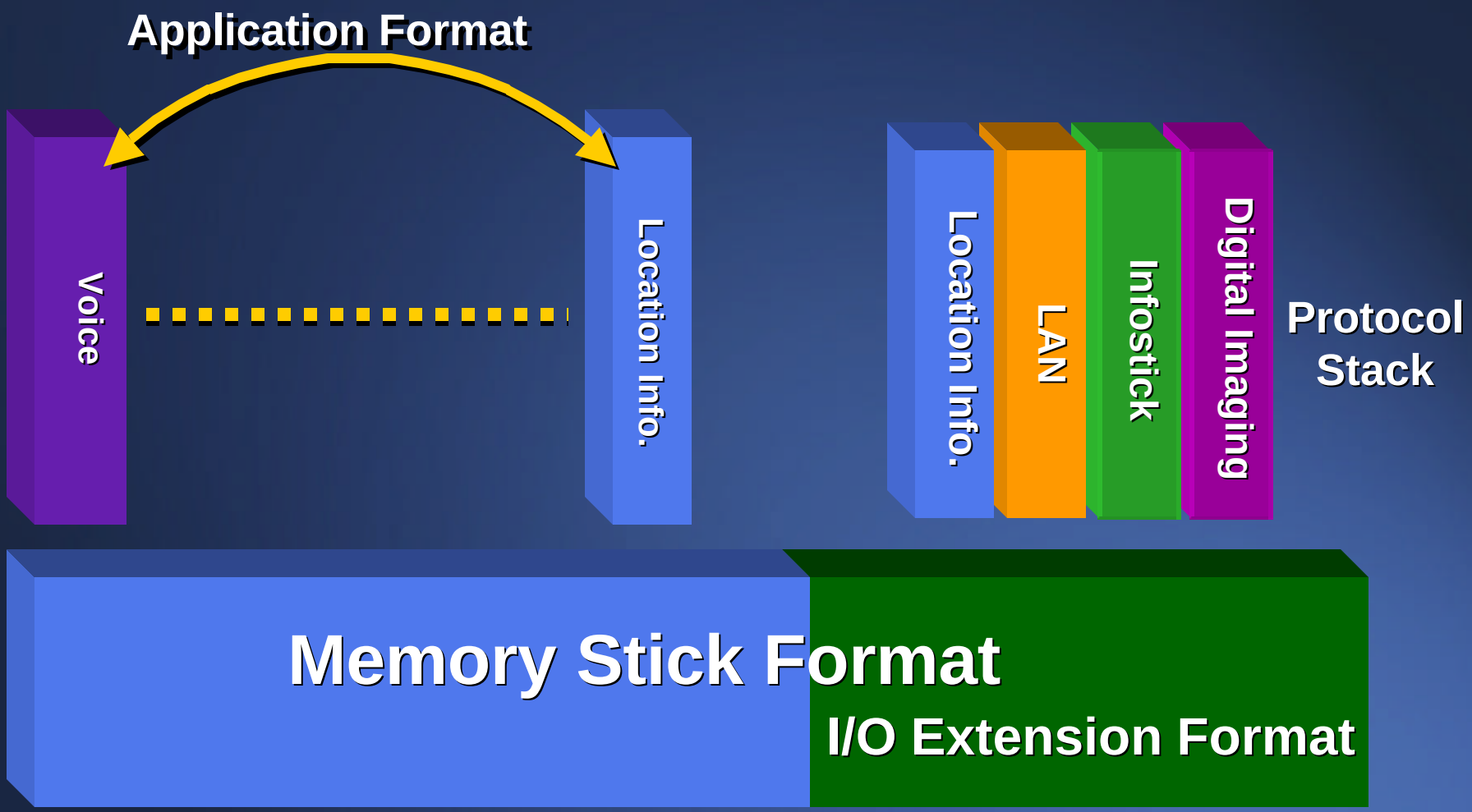
SONY



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I/O Extension Format Concept



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I/O Extension Basic Concept

Extend Registers for Various Functions
Leverage Interoperability of "Memory Stick Platform"

- Follow the original electrical specification
- Use same TPC Code
  (Transfer Protocol Command)
- Shall extend additional functionality
- The data transfer direction should be the same
- Accessible memory should be less than 512KB
  - Avoid function overlapping with existing Memory Stick

Memory Stick Information Available

SONY

For Making Memory Stick Expansion Modules, Visit Us At…

- Memory Stick Information for Developers
  - http://www.memorystick.org/e-index.html
    - Concept>License Agreement>Marketing Information
- What is Memory Stick Expansion Module?
  - http://www.memorystick.org/msinfo/eng/io/index.html
- Memory Stick Home page
  - http://www.world.sony.com/Electronics/MS/Index.html
    - About Memory Stick

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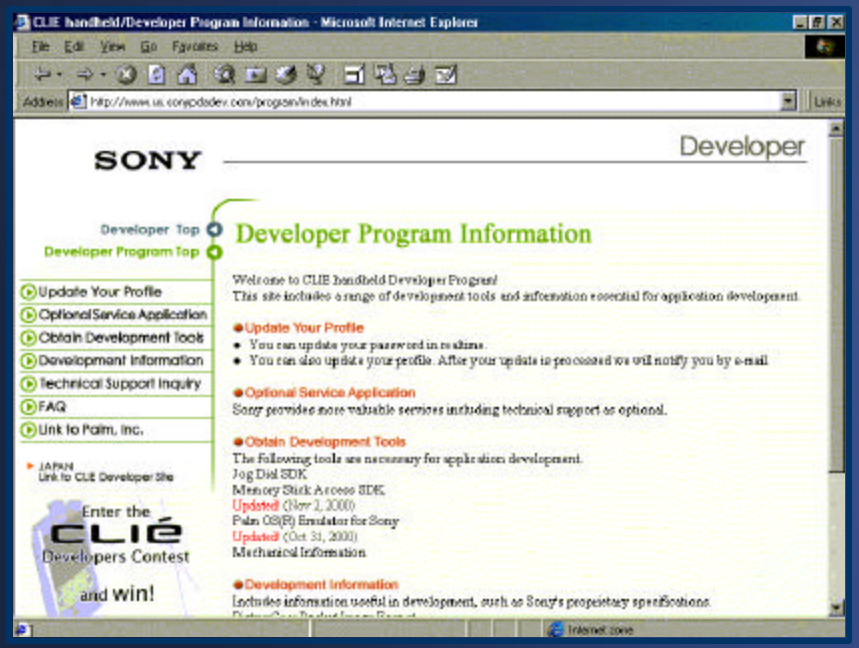
CLIÉ Handheld Tools Ready for Download

SONY

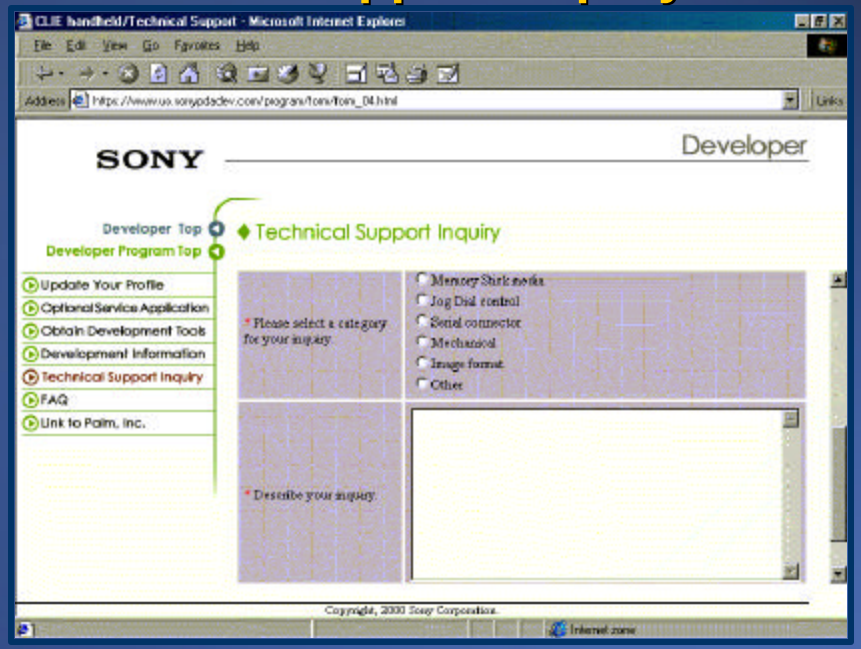
Visit the CLIÉ Handheld Developer Site At…

https://www.us.sonypdadev.com/program/

Developer Program Information



Technical Support Inquiry Form



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We Do Our Best to Meet YOUR Demands!

SONY

Send Us Your Comments Through Jan. 15, 2001

Mail to:expstick@ita.sel.sony.com

Please let us know…

- What are you planning to do?
- What will you need from us?

The Plans for YOU Will Be Announced in Feb.

On the CLIÉ Handheld Developer Site…

http://www.us.sonypdadev.com/

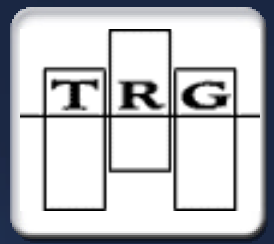
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TRG

Mark Kubovich
VP of Operations

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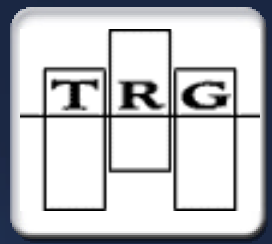
Agenda



- Introduction to TRGpro's expansion slot
- CompactFlash Overview
- CompactFlash Specifications
- Listing of Cards
- Palm OS® Support for CompactFlash
- Development Tools Available
- Future Expansion Solutions

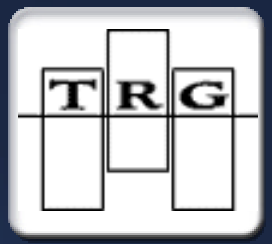
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Introduction to TRGpro's Expansion



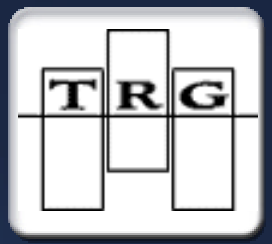
- TRGpro available in January, 2000
- Selling primarily to enterprise clients
- Utilizes CompactFlash expansion media
- Other expansion media were considered
  - PCMCIA
  - SmartMedia Card
  - MultiMedia Card
  - MemoryStick
  - Proprietary

CompactFlash Overview



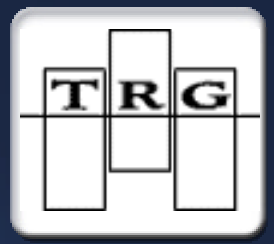
- Mature industry standard (CFA founded in 1995, 143 member companies including TRG and 3COM)
- Growing list of vendors and functions
- Cards can be interchanged with other systems
- Small enough to fit Palm form factor
- Contains ATA interface to facilitate use of FAT File System (FFS) storage cards

CompactFlash Specifications



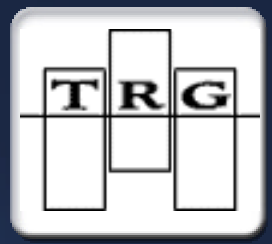
- TRGpro supports CF v1.4, Power Level 0, Type I and Type II
- Physical Size: 36.4 mm X 42.8 mm X 5 mm
- 16-bit, device-independent interface
- Up to 400 mA peak current draw

Cards Currently Available



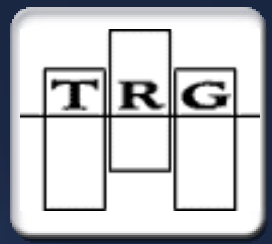
- Storage
  - Numerous manufacturers including SanDisk, IBM
  - Capacity ranges from 8 MB to 1 GB
- Connectivity
  - Serial, Digital phone cards, Ethernet, Bluetooth, CDPD
- Productivity
  - Bar code laser scanners, wand scanners, GPS

Palm OS® Support for CompactFlash



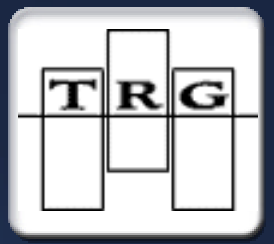
- Accomplished through extensions
- TRG supplies drivers for certain cards:
  - All storage cards
  - UART interface
  - Ethernet
- OS Extensions provide standard software interface to card slot
- Will support Suspenders after release of OS 4.0

Development Tools and Support



- Hardware Development Kit
  - Includes description of all relevant hardware information
- Software Development Kit
  - Documents TRG's Palm OS extensions, and provides sample code
- CompactFlash Association (CFA)
  - Responsible for maintaining and promoting the CF specification
  - Most current information requires membership

Future Expansion Solutions



- Multiple expansion slots on a single Palm device
- Multiple expansion media on a single Palm device
- Access standards are needed to ease software development

More Info…

- Web: www.palmos.com
- Email: developer_alliances@palm.com

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